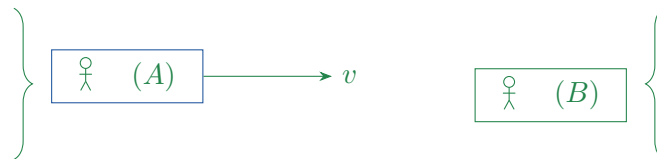


1. Introduction to STR

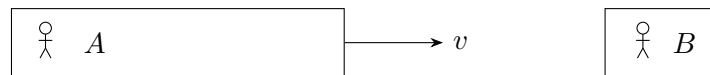
Sunday, 2 May 2021 9:08 PM

1 Introduction to STR



We can't possibly tell who is moving.

accelerated frame can be detected but inertial frames can't.



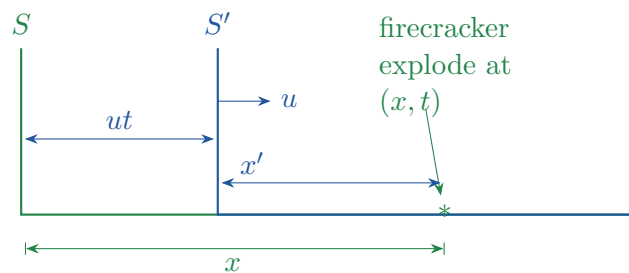
$$\vec{v}_{AB} = v \hat{i}$$

$$\vec{v}_{BA} = \vec{v}_B - \vec{v}_A = v(-\hat{i})$$

To say everything is same in both frames $\Rightarrow$ laws of newton are same.

Event

Event :— Something happens at some time 't' at some position 'x'.



An event occurs at (x, t) in frame S .

Same event occurs at (x', t') in frame S' .

When both origin coincided, clocks were synchronised.

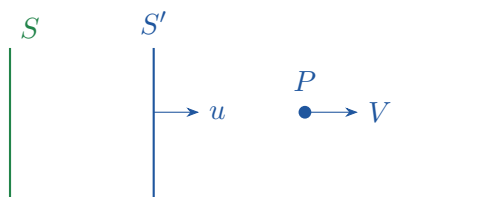
Galilean transformation

We can see that,

$$\begin{aligned} x' &= x - ut \\ y' &= y \\ z' &= z \\ t' &= t \end{aligned}$$

Time is just time in Galilean transformation.

2 Consequences of Galilean transformation :-



let object is moving with speed

$$v = \frac{\Delta x}{\Delta t}$$

Velocity of object according to S'

$$= \frac{\Delta x'}{\Delta t'} = \frac{(\Delta x - u \Delta t)}{\Delta t}$$

$$v' = \left(\frac{\Delta x}{\Delta t} - u \right)$$

$$\boxed{v' = v - u}$$

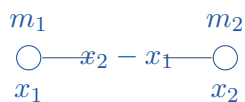
Accelerations are still same

$$a' = \frac{dv'}{dt} = 0$$

$$\text{also } a = \frac{dv}{dt} = 0$$

Two bodies under mutual force

let us consider two bodies under mutual force



For frame S :-

$$F_1 = m a_1 \Rightarrow \frac{G m_1 m_2}{(x_1 - x_2)^2} = m_1 a_1$$

$$a_1 = \frac{G m_2}{(x_1 - x_2)^2} \hat{i}$$

$$F_2 = m a_2 \Rightarrow \frac{G m_1 m_2}{(x_1 - x_2)^2} = m_2 a_2$$

$$a_2 = \frac{G m_1}{(x_1 - x_2)^2} (-\hat{i})$$

For frame S') :-

$$F'_1 = m_1 a'_1 \Rightarrow \frac{G m_1 m_2}{(x'_1 - x'_2)^2} = m_1 a'_1$$

$$a'_1 = \frac{G m_2}{(x'_1 - x'_2)^2} = \frac{G m_2}{(x_1 - vt - x_2 + vt)^2}$$

$$a'_1 = \frac{G m_2}{(x_1 - x_2)^2} \hat{i} = a_1$$

Similarly

$a'_2 = a_2$

3 300 Years later :-

E-M waves have been identified with light or vice versa. with speed

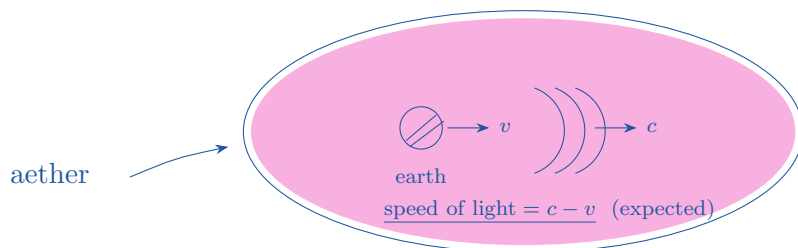
$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = 3 \times 10^8 \text{ m/s}$$

In general waves moves w.r.t medium so, what is the medium which carries the waves of light (EM).

The medium should be everywhere (as it travels through so much space from stars as well).

We know that to have such large speed, we need very dense medium, planets have been moving around for ages but has not been slowed down, so it is a peculiar medium.

The aether



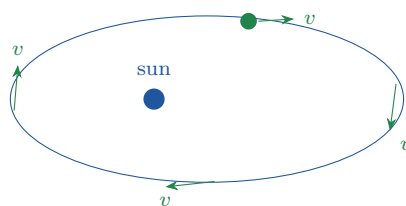
But experimentally speed came out to be 'c'.

So, people thought velocity of earth w.r.t aether = 0

$$\text{So Speed of light} = c - 0 = c$$

But how can on each day while earth is revolving around sun, and on each day we get speed of light = c i.e. ($V_e \rightarrow 0$) But it is impractical!

$$V_e \neq 0 \text{ on each day}$$



(So, Aether does not exist) And speed of light is same in each inertial frames of reference.

If speed of light depend upon how fast I move, then It is possible to detect speed of observer (self) and that would mean there exist a fixed / preferred frame of reference.

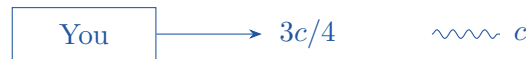
Einstien postulates :-

So Einstien postulates :-

- (1) All laws of nature are equivalent in all frames of reference (All Inertial observers are equivalent)
- (2) Speed of light is const in all inertial frames.

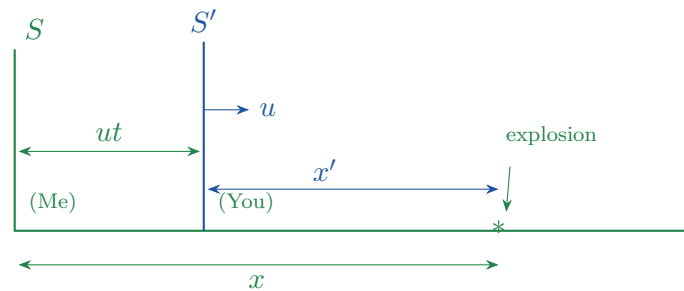
we / scientists believe that all laws of nature behave the same in all frames of reference.

Imagine –



you should get speed $c/4$ but you keep getting 'c', so, You are probably using shorter scales to measure. or time in your frame is moving slower (You kept light running for 4 sec & calling it one sec).

4 Lorentz Transformations :-



I will find coordinate of event according to you as $(x - ut)$. I know you are moving so your meter sticks have shrunk and thus will measure longer x . So I multiply by some factor ' γ ' to get same result of x' .

$$x' = (x - ut) \gamma \quad \text{--- ①}$$

Similarly, you find $(x = x' + ut')$ But you know I am moving, so I will be using short meter sticks and thus my measurements $(x > x' + ut')$. So To match up with that you can predict —

$$x = (x' + ut') \gamma \quad \text{--- ②}$$

Multiply ① & ②

$$xx' = \gamma^2 (xx' + xut' - utx' - u^2tt')$$

$$1 = \gamma^2 \left(1 + \frac{ut'}{x'} - \frac{ut}{x} - u^2 \frac{t}{x} \frac{t'}{x'} \right)$$

Also as both origins coincided, A light pulse travels to right which sets up explosion

$$x = ct \quad \& \quad x' = ct'$$

$$\Rightarrow 1 = \gamma^2 \left(1 + \frac{u}{c} - \frac{u}{c} - \frac{u^2}{c^2} \right)$$

$$\Rightarrow \gamma = \frac{1}{\sqrt{1 - u^2/c^2}}$$

$$\Rightarrow \boxed{\begin{cases} x' = \frac{(x - ut)}{\sqrt{1 - u^2/c^2}} \\ x = \frac{x' + ut'}{\sqrt{1 - u^2/c^2}} \end{cases} \quad \begin{cases} t' = \frac{x'}{c} = \left(\frac{x}{c} - \frac{u t}{c} \right) \gamma \\ t' = \left(t - \frac{u x}{c^2} \right) \gamma \end{cases}}$$

So (x, t) is coordinate of event E : Fire cracker explosion in frame S

$\hookrightarrow$ same event E has coordinate (x', t') for frame S' .